



**SOLID**Proof  
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**Blockchain Security | Smart Contract Audits | KYC  
Development | Marketing**

MADE IN GERMANY

**Satisfy  
Network  
AUDIT**

**04. July, 2024**

**FOR**



**SolidProof.io**



**@solidproof.io**



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## Introduction

[SolidProof.io](#) is a brand of the officially registered company MAKE Network GmbH, based in Germany. We're mainly focused on Blockchain Security such as Smart Contract Audits and KYC verification for project teams.

Solidproof.io assess potential security issues in the smart contracts implementations, review for potential inconsistencies between the code base and the whitepaper/documentation, and provide suggestions for improvement.

## Disclaimer

[SolidProof.io](#) reports are not, nor should be considered, an "endorsement" or "disapproval" of any particular project or team. These reports are not, nor should be considered, an indication of the economics or value of any "product" or "asset" created by any team. SolidProof.io do not cover testing or auditing the integration with external contract or services (such as Unicrypt, Uniswap, PancakeSwap etc'...)

**SolidProof.io Audits do not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technology proprietors. SolidProof Audits should not be used in any way to make decisions around investment or involvement with any particular project. These reports in no way provide investment advice, nor should be leveraged as investment advice of any sort.**

SolidProof.io Reports represent an extensive auditing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology. Blockchain technology and cryptographic assets present a high level of ongoing risk. SolidProof's position is that each company and individual are responsible for their own due diligence and continuous security. SolidProof in no way claims any guarantee of the security or functionality of the technology we agree to analyze.

## Project Overview

### Summary

| Project Name      | Satisfy Network                                                                                                                                                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Website           | <a href="https://satisfy.network">https://satisfy.network</a>                                                                                                                                                                                                             |
| About the project | <b>Satisfy Netowrk</b> is a decentralized funding protocol for Web3. It personalized digital funding on BSC blockchain, seamlessly merging AI with blockchain. It aligns stakeholder incentives during and beyond the fundraiser, Finally makes everyone to be satisfied. |
| Chain             | TBA                                                                                                                                                                                                                                                                       |
| Language          | Solidity                                                                                                                                                                                                                                                                  |
| Codebase          | <a href="https://bscscan.com/address0xbd4eb6bdde80b3d39c55c52c91cc9c638b8055ea#code">https://bscscan.com/address0xbd4eb6bdde80b3d39c55c52c91cc9c638b8055ea#code</a>                                                                                                       |
| Commit            | N/A                                                                                                                                                                                                                                                                       |

### Social Medias

| Telegram  | <a href="https://t.me/satisfyNetwork">https://t.me/satisfyNetwork</a>             |
|-----------|-----------------------------------------------------------------------------------|
| Twitter   | N/A                                                                               |
| Facebook  | N/A                                                                               |
| Instagram | N/A                                                                               |
| GitHub    | <a href="https://github.com/SatisfyNetwork">https://github.com/SatisfyNetwork</a> |
| Reddit    | N/A                                                                               |
| Medium    | N/A                                                                               |
| Discord   | N/A                                                                               |
| YouTube   | N/A                                                                               |
| TikTok    | N/A                                                                               |
| LinkedIn  | N/A                                                                               |

## Audit Summary

| Version | Delivery Date | Change Log                                                                                                                    |
|---------|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| v1.0    | 4. July 2024  | <ul style="list-style-type: none"> <li>Layout Project</li> <li>Automated/ Manual-Security Testing</li> <li>Summary</li> </ul> |

**Note** - This Audit report consists of a security analysis of the **Satisfy Network** project. This analysis did not include functional testing (or unit testing) of the contract's logic.



## File Overview

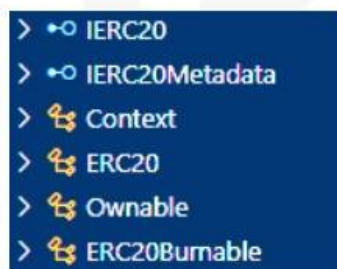
The Team provided us with the files that should be tested in the security assessment. This audit covered the following files listed below with an SHA-1 Hash.

| File Name                    | SHA-1 Hash                               |
|------------------------------|------------------------------------------|
| contracts/SatisfyNetwork.sol | aac3fd55f18d3a880c11c1481b780ed70fb11648 |

*Please note: Files with a different hash value than in this table have been modified after the security check, either intentionally or unintentionally. A different hash value may (but need not) be an indication of a changed state or potential vulnerability that was not the subject of this scan.*

## Imported packages

*Used code from other Frameworks/Smart Contracts (direct imports).*



**Note for Investors:** We only Audited a simple token contract for **Satisfy Network**. However, If the project has other contracts (for example, a Presalecontract etc), and they were not provided to us in the audit scope, then we cannot comment on its security and are not responsible for it in any way.

## Audit Information

### Vulnerability & Risk Level

Risk represents the probability that a certain source threat will exploit the vulnerability and the impact of that event on the organization or system. The risk level is computed based on CVSS version 3.0.

| Level                | Value   | Vulnerability                                                                                                                               | Risk (Required Action)                                              |
|----------------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Critical</b>      | 9 - 10  | A vulnerability that can disrupt the contract functioning in a number of scenarios, or creates a risk that the contract may be broken.      | Immediate action to reduce risk level.                              |
| <b>High</b>          | 7 - 8.9 | A vulnerability that affects the desired outcome when using a contract, or provides the opportunity to use a contract in an unintended way. | Implementation of corrective actions as soon as possible.           |
| <b>Medium</b>        | 4 - 6.9 | A vulnerability that could affect the desired outcome of executing the contract in a specific scenario.                                     | Implementation of corrective actions in a certain period.           |
| <b>Low</b>           | 2 - 3.9 | A vulnerability that does not have a significant impact on possible scenarios for the use of the contract and is probably subjective.       | Implementation of certain corrective actions or accepting the risk. |
| <b>Informational</b> | 0 - 1.9 | A vulnerability that have informational character but is not effecting any of the code.                                                     | An observation that does not determine a level of risk              |

## Auditing Strategy and Techniques Applied

Throughout the review process, care was taken to check the repository for security-related issues, code quality, and compliance with specifications and best practices. To this end, our team of experienced pen-testers and smart contract developers reviewed the code line by line and documented any issues discovered.

We check every file manually. We use automated tools only so that they help us achieve faster and better results.

## Methodology

The auditing process follows a routine series of steps:

1. Code review that includes the following:
  - a. Reviewing the specifications, sources, and instructions provided to SolidProof to ensure we understand the size, scope, and functionality of the smart contract.
  - b. Manual review of the code, i.e., reading the source code line by line to identify potential vulnerabilities.
  - c. Comparison to the specification, i.e., verifying that the code does what is described in the specifications, sources, and instructions provided to SolidProof.
2. Testing and automated analysis that includes the following:
  - a. Test coverage analysis determines whether test cases cover code and how much code is executed when those test cases are executed.
  - b. Symbolic execution, which is analysing a program to determine what inputs cause each part of a program to execute.
3. Review best practices, i.e., review smart contracts to improve efficiency, effectiveness, clarity, maintainability, security, and control based on best practices, recommendations, and research from industry and academia.
4. Concrete, itemized and actionable recommendations to help you secure your smart contracts.



## Overall Security

### Medium or higher issues

**critical Issues found**

**✗ Contract is not safe to deploy**

Description

The contract does contain issues of high or medium criticality. This means that known vulnerabilities were found in the source code and should be fixed asap.

Comment

Here, Owner is able to lock the transfer and blacklist wallets that can block the user to transfer tokens.



## Upgradeability

### Contract is not an upgradable

✓ **Deployer cannot update the contract with new functionalities.**

Description

The contract is not an upgradeable contract. The Deployer is not able to change or add any functionalities to the contract after deploying.

Comment

N/A



## Ownership

The ownership is not renounced

**X** The ownership is not renounced

Description

The owner has not renounced the ownership that means that the owner retains control over the contract's operations, including the ability to execute functions that may impact the contract's users or stakeholders. This can lead to several potential issues, including:

- Centralizations
- The owner has significant control over contract's operations.

Example

N/A

Comment

N/A

**Note** – *The contract cannot be considered as renounced till it is not deployed or having some functionality that can change the state of the contract.*

## Ownership Privileges

*These functions can be dangerous. Please note that abuse can lead to financial loss. We have a guide where you can learn more about these Functions.*

### Minting tokens

*Minting tokens refer to the process of creating new tokens in a cryptocurrency or blockchain network. This process is typically performed by the project's owner or designated authority, who has the ability to add new tokens to the network's total supply.*

**Contract owner cannot mint new tokens**

 **The owner cannot mint new tokens**

|             |                                                                         |
|-------------|-------------------------------------------------------------------------|
| Description | The owner is not able to mint new tokens once the contract is deployed. |
| Comment     | N/A                                                                     |



## Burning tokens

*Burning tokens is the process of permanently destroying a certain number of tokens, reducing the total supply of a cryptocurrency or token. This is usually done to increase the value of the remaining tokens, as the reduced supply can create scarcity and potentially drive up demand.*

**Contract owner cannot burn tokens**



**The owner cannot burn tokens**

Description

The owner is not able burn tokens without any allowances.

Comment

N/A



## Blacklist addresses

*Blacklisting addresses in smart contracts is the process of adding a certain address to a blacklist, effectively preventing them from accessing or participating in certain functionalities or transactions within the contract. This can be useful in preventing fraudulent or malicious activities, such as hacking attempts or money laundering.*

**owner cannot blacklist address.**

 **The owner cannot blacklist addresses**

Description

The owner cannot blacklist addresses from transferring of tokens.

Comment

N/A

## Fees and Tax

*In some smart contracts, the owner or creator of the contract can set fees for certain actions or operations within the contract. These fees can be used to cover the cost of running the contract, such as paying for gas fees or compensating the contract's owner for their time and effort in developing and maintaining the contract.*

**Contract owner cannot set fees more than 25%**

☒ **The owner cannot set Fees more than 25%.**

|             |                                                  |
|-------------|--------------------------------------------------|
| Description | The owner is not able to set the fees above 25%. |
| Comment     | N/A                                              |

## Lock User Funds

*In a smart contract, locking refers to the process of restricting access to certain tokens or assets for a specified period of time. When token or assets are locked in a smart contract, they cannot be transferred or used until the lock-up period has expired or certain conditions have been met.*

**Contract owner can lock the user funds**

**✗ The owner is able to lock the contract**

Description

Locking the contract means that the owner is able to lock any funds of addresses that they are not able to transfer bought tokens anymore.

Comment

Here, The blockTransfer variable can be able to lock the transfer of the tokens till 01Jan, 2025.

**File, Line/s: L629-631**

**Codebase:**

```
function setBlockTransfer(bool _bool1) public onlyOwner {
    blockTransfer = _bool1;
}
```

## External/Public functions

External/public functions are functions that can be called from outside of a contract, i.e., they can be accessed by other contracts or external accounts on the blockchain. These functions are specified using the function declaration's external or public visibility modifier.

## State variables

State variables are variables that are stored on the blockchain as part of the contract's state. They are declared at the contract level and can be accessed and modified by any function within the contract. State variables can be needed within visibility modifier, such as public, private or internal, which determines the access level of the variable.

## Components

|                                                                                                    |                                                                                                    |                                                                                                     |                                                                                                   |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  <b>Contracts</b> |  <b>Libraries</b> |  <b>Interfaces</b> |  <b>Abstract</b> |
| 2                                                                                                  | 0                                                                                                  | 2                                                                                                   | 3                                                                                                 |

## Exposed Functions

This section lists functions that are explicitly declared public or payable. Please note that getter methods for public stateVars are not included.

|                                                                                                   |                                                                                                    |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
|  <b>Public</b> |  <b>Payable</b> |
| 29                                                                                                | 0                                                                                                  |

| External | Internal | Private | Pure | View |
|----------|----------|---------|------|------|
| 9        | 38       | 0       | 0    | 16   |

## StateVariables

|              |                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------|
| <b>Total</b> |  <b>Public</b> |
| 9            | 3                                                                                                 |



## Capabilities

| Solidity Versions observed |  Experimental Features |  Can Receive Funds |  Uses Assembly |  Has Destroyable Contracts |
|----------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| ^0.8.18                    | _____                                                                                                   | _____                                                                                               | _____                                                                                            | _____                                                                                                         |





## Inheritance Graph

*An inheritance graph is a graphical representation of the inheritance hierarchy among contracts. In object-oriented programming, inheritance is a mechanism that allows one class (or contract, in the case of Solidity) to inherit properties and methods from another class. It shows the relationships between different contracts and how they are related to each other through inheritance.*



## Centralization Privileges

*Centralization can arise when one or more parties have privileged access or control over the contract's functionality, data, or decision-making. This can occur, for example, if the contract is controlled by a single entity or if certain participants have special permissions or abilities that others do not.*

In the project, there are authorities that have access to the following functions:

| File                      | Privileges                                                                                                                                                                       |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SatisfyNetwork.sol</b> | <ul style="list-style-type: none"> <li>➤ The owner can whitelist wallets.</li> <li>➤ The owner can lock transfer by setting blockTransfer to true till 01. Jan, 2025.</li> </ul> |

## Recommendations

To avoid potential hacking risks, it is advisable for the client to manage the private key of the privileged account with care. Additionally, we recommend enhancing the security practices of centralized privileges or roles in the protocol through a decentralized mechanism or smart-contract-based accounts, such as multi-signature wallets.

Here are some suggestions of what the client can do:

- Consider using multi-signature wallets: Multi-signature wallets require multiple parties to sign off on a transaction before it can be executed, providing an extra layer of security e.g. Gnosis Safe
- Use of a timelock at least with a latency of e.g. 48-72 hours for awareness of privileged operations
- Introduce a DAO/Governance/Voting module to increase transparency and user involvement
- Consider Renouncing the ownership so that the owner cannot modify any state variables of the contract anymore. Make sure to set up everything before renouncing.

## Audit Result

### #1| Owner can Lock Transfer

| File               | Severity | Location | Status |
|--------------------|----------|----------|--------|
| SatisfyNetwork.sol | Medium   | L629-631 | ACK    |

#### Description

- The owner can set blockTransfer variable to true and is able to lock the user's funds, only whitelist addresses can be able to transfer tokens till 01. Jan 2025 after that no locking functionality will remain in this contract.

### #2| Local variables shadowing(shadowing- local)

| File               | Severity | Location | Status |
|--------------------|----------|----------|--------|
| SatisfyNetwork.sol | Low      | L600,607 | ACK    |

#### Description

- Rename the local variables that shadow another component.

### #3| Missing events Arithmetic(events-maths)

| File               | Severity | Location | Status |
|--------------------|----------|----------|--------|
| SatisfyNetwork.sol | Low      | L629-631 | ACK    |

#### Description

- Emit an event for critical parameter changes.

### #4| Floating Pragma solidity version

| File               | Severity      | Location | Status |
|--------------------|---------------|----------|--------|
| SatisfyNetwork.sol | Informational | L2       | ACK    |

#### Description



- The contracts use floating pragma solidity versions which are not recommended for deployment as they may be susceptible to known vulnerabilities.

## #5| NatSpec Documentation missing

| File               | Severity      | Location | Status |
|--------------------|---------------|----------|--------|
| SatisfyNetwork.sol | Informational | --       | ACK    |

### Description

- If you started to comment on your code, also comment on all other functions, variables, etc.



## Legend for the Issue Status

| Attribute or Symbol      | Meaning                                                                |
|--------------------------|------------------------------------------------------------------------|
| <b>Open</b>              | The issue is not fixed by the project team.                            |
| <b>Fixed</b>             | The issue is fixed by the project team.                                |
| <b>Acknowledged(ACK)</b> | The issue has been acknowledged or declared as part of business logic. |





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